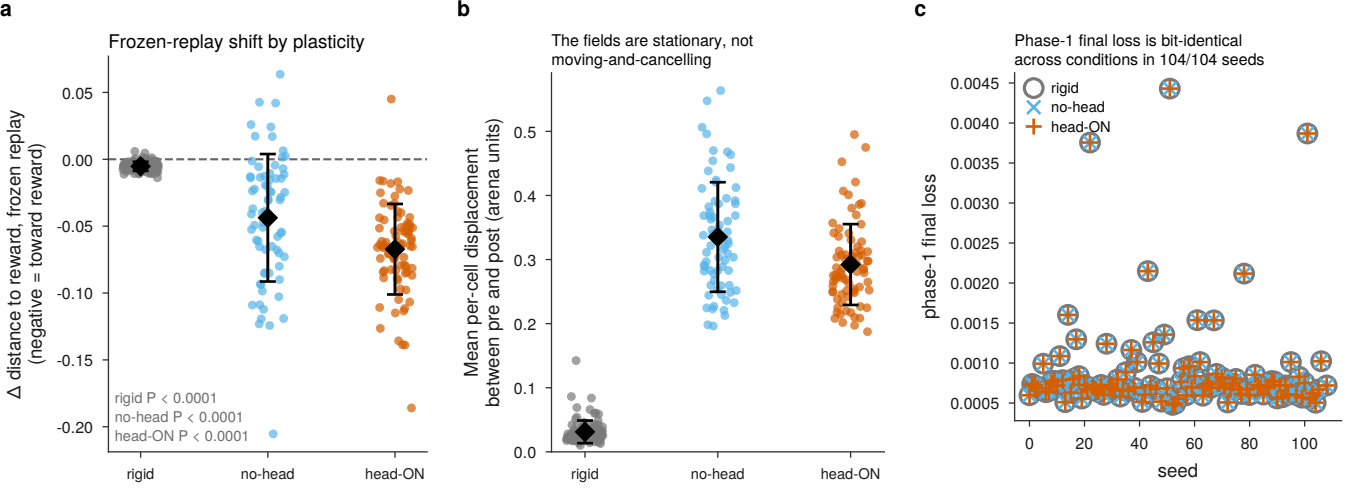

a
Frozen-replay shift by plasticity
Δ distance to reward, frozen replay
(negative = toward reward)
0.05
0.00
-0.05
-0.10
-0.15
-0.20
rigid P < 0.0001
no-head P < 0.0001
head-ON P < 0.0001
rigid
no-head
head-ON
b
The fields are stationary, not
moving-and-cancelling
Mean per-cell displacement
between pre and post (arena units)
0.5
0.4
0.3
0.2
0.1
0.0
rigid
no-head
head-ON
c
Phase-1 final loss is bit-identical
across conditions in 104/104 seeds
rigid
no-head
head-ON
phase-1 final loss
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0
20
40
60
80
100
seed